

Manuscripts considered — cca-mets-scope-unknown-z3p9

Master inventory: every paper reviewed in this case

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Manuscripts considered — cca-mets-scope-unknown-z3p9

Master inventory: 16 clinical (12 included, 4 considered & excluded) + 12 pre-clinical (10 included, 2 considered & excluded) + 4 additional trial publications/registrations. One entry per manuscript, sorted by year (newest first). Excluded entries are papers a Libby agent reviewed and chose NOT to feed to the board, with the exclusion reason captured.

Meric-Bernstam/Raghav (2024) — Journal of Clinical Oncology

Reference: [PMID 37870536](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Trastuzumab deruxtecan (T-DXd)
- **Indication / model:** HER2-expressing biliary tract cancer (pan-tumor basket cohort) (2L+); Biliary tract cohort of a HER2-expressing solid-tumor basket; central HER2 IHC 3+ or 2+. IHC 3+ subgroup responded markedly better than the overall cohort
- **Design:** Open-label, multicohort phase 2 basket (DESTINY-PanTumor02)
- **n:** 41
- **Effect size:** 22.0 % — ORR, biliary tract cohort (investigator)
- **Variance:** 95% CI 10.6–37.6
- **Toxicities:** Drug-related grade 3+ AE (any) $G \geq 3$ (40.8%); ILD/pneumonitis (10.5%); ILD/pneumonitis G5; Nausea
- **Case fit:** partial
- **Notes:** Biomarker-defined pan-tumor basket; biliary patients enroll via a HER2-expressing cohort, so the IHC score drives expected benefit. ILD is the defining safety concern. AE rates are study-wide, not biliary-cohort-specific, per the JCO report.

—/— (2024)

Reference: — · **Type:** clinical · **Inclusion:** excluded — Phase 1 dose-escalation (NCT06160752) with no published efficacy or safety readout yet. Next-generation FGFR2 inhibitor relevant if the patient is FGFR2-positive and progresses on pemigatinib/futibatinib, but there is no clinical-evidence paper to compile.

- **Intervention:** TYRA-200
- **Notes:** Excluded: Phase 1 dose-escalation (NCT06160752) with no published efficacy or safety readout yet. Next-generation FGFR2 inhibitor relevant if the patient is FGFR2-positive and progresses on pemigatinib/futibatinib, but there is no clinical-evidence paper to compile.

—/— (2024)

Reference: — · **Type:** clinical · **Inclusion:** excluded — Ongoing first-line phase 3 (NCT06467357) with no readout. The T-DXd evidence base is already captured from DESTINY-PanTumor02; this row would add a registration-only entry with no published outcomes.

- **Intervention:** Trastuzumab deruxtecan + rilvegostomig
- **Notes:** Excluded: Ongoing first-line phase 3 (NCT06467357) with no readout. The T-DXd evidence base is already captured from DESTINY-PanTumor02; this row would add a registration-only entry with no published outcomes.

—/— (2024)

Reference: — · **Type:** clinical · **Inclusion:** excluded — Ongoing first-line phase 3 (NCT06282575) with no readout. The zanidatamab evidence base is captured from HERIZON-BTC-01; nothing published to extract here.

- **Intervention:** Zanidatamab + chemo +/- checkpoint
- **Notes:** Excluded: Ongoing first-line phase 3 (NCT06282575) with no readout. The zanidatamab evidence base is captured from HERIZON-BTC-01; nothing published to extract here.

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** TYRA-200 (next-generation FGFR2 inhibitor)
- **Indication / model:** Advanced cholangiocarcinoma with activating FGFR2 alterations (2L+); Activating FGFR2 gene alteration (incl. acquired resistance mutations)
- **Design:** Open-label phase 1 dose-escalation/expansion
- **Effect size:** — — Safety, tolerability, recommended phase 2 dose; antitumor activity
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Trastuzumab deruxtecan + rilvegostomig (anti-PD-1/TIGIT bispecific)
- **Indication / model:** First-line advanced HER2-expressing biliary tract cancer (1L); HER2-expressing (IHC)
- **Design:** Randomized phase 3, T-DXd + rilvegostomig vs standard of care
- **Effect size:** — — Progression-free survival / overall survival
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

—/— (2024)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Zanidatamab + cisplatin/gemcitabine +/- pembrolizumab or durvalumab
- **Indication / model:** First-line advanced HER2-positive biliary tract cancer (1L); HER2-positive (IHC/ISH)
- **Design:** Randomized phase 3, zanidatamab + chemo +/- checkpoint vs chemo
- **Effect size:** — — Progression-free survival / overall survival
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Goyal/Bridgewater (2023) — New England Journal of Medicine

Reference: [PMID 36652354](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Futibatinib
- **Indication / model:** Previously treated advanced intrahepatic cholangiocarcinoma with FGFR2 fusion or

rearrangement (2L+); Centrally confirmed FGFR2 fusion/rearrangement in intrahepatic CCA, ≥ 1 prior line; activity retained against some pemigatinib-resistance kinase-domain mutations

- **Design:** Single-arm, open-label phase 2 (FOENIX-CCA2)
- **n:** 103
- **Effect size:** 42.0 % — ORR (independent central review)
- **Variance:** 95% CI 32.0–52.0
- **Toxicities:** Hyperphosphatemia $G \geq 3$ (n=103, 30%); Increased AST $G \geq 3$ (n=103, 7%); Stomatitis $G \geq 3$ (n=103, 6%); Fatigue $G \geq 3$ (n=103, 6%); Treatment-related discontinuation (n=103, 2%)
- **Case fit:** partial
- **Notes:** Irreversible FGFR1-4 inhibitor with activity against some acquired FGFR2 kinase-domain resistance mutations, relevant for sequencing after pemigatinib. Efficacy from NEJM full text; toxicities are the grade 3+ rates reported in the abstract/results, full per-term table not extracted.

Harding/Bekaii-Saab (2023) — Lancet Oncology

Reference: [PMID 37423227](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Zanidatamab
- **Indication / model:** Previously treated HER2-amplified biliary tract cancer (2L+); Cohort 1: HER2 amplified by ISH plus IHC 2+/3+ (4B5 companion antibody); prior gemcitabine-based therapy
- **Design:** Single-arm, open-label phase 2b (HERIZON-BTC-01)
- **n:** 80
- **Effect size:** 41.3 % — Confirmed ORR (independent central review)
- **Variance:** 95% CI 30.4–52.8
- **Toxicities:** Diarrhea (n=80, 40%); Diarrhea $G3$ (4/80, 5%); Infusion-related reaction (n=80, 35%); Decreased ejection fraction $G3$ (3/80, 3%)
- **Case fit:** partial
- **Notes:** Pivotal trial driving FDA accelerated approval of zanidatamab in HER2-positive (IHC 3+) BTC. Gating test is HER2 IHC with reflex ISH, not NGS alone, and is untested here. Efficacy/safety from Lancet Oncology abstract and conference follow-up.

Bekaii-Saab/Pant (2023) — Journal of Clinical Oncology

Reference: [PMID 37099736](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Adagrasib
- **Indication / model:** KRAS G12C-mutated advanced solid tumors, biliary tract cancer cohort (2L+); KRAS G12C-mutant pretreated GI tumors; biliary tract cancer cohort n=12 within a pan-tumor basket
- **Design:** Open-label phase 1/2 basket (KRYSTAL-1), pan-tumor cohort
- **n:** 12
- **Effect size:** 41.7 % — ORR, biliary tract cohort
- **Toxicities:** Treatment-related grade 3-4 AE (any) $G \geq 3$ (n=63, 27%); Nausea (n=63, 49.2%); Diarrhea (n=63, 47.6%); Fatigue (n=63, 41.3%); Vomiting (n=63, 39.7%)
- **Case fit:** partial
- **Notes:** Published KRYSTAL-1 biliary cohort (n=12) gives a small but on-indication KRAS G12C readout, supersedes the no-readout basket NCT05722327 on trials.jsonl. KRAS G12C is rare in CCA (~1%) and gated on tumor NGS.

—/— (2023)

Reference: — · **Type:** clinical · **Inclusion:** excluded — Registry-only basket entry with no standalone readout; superseded by the published KRYSTAL-1 biliary cohort (PMID 37099736), which is compiled as an included row.

- **Intervention:** Adagrasib (basket NCT05722327)
- **Notes:** Excluded: Registry-only basket entry with no standalone readout; superseded by the published KRYSTAL-1 biliary cohort (PMID 37099736), which is compiled as an included row.

Weisser/Dixit (2023) — Nature Communications

Reference: [PMID 36914633](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Zanidatamab (ZW25)
- **Indication / model:** HER2-amplified gastric models: NCI-N87 cell-line xenograft and GXA 3054 patient-derived xenograft
- **Design:** Biparatopic antibody binding the ECD2 (pertuzumab) and ECD4 (trastuzumab) epitopes on adjacent HER2 molecules, which forces HER2 into large surface clusters. The clustering drives receptor internalization, blocks ligand-dependent and independent signaling, and uniquely triggers complement-dependent cytotoxicity alongside ADCC and ADCP.
- **n:** not reported per arm
- **Effect size:** strong — Zanidatamab outperformed both trastuzumab and the trastuzumab-pertuzumab pair, with 268% growth-rate inhibition at 4 mg/kg in NCI-N87 (versus 86% and 155%) and tumor regression in the high-HER2 PDX. The CDC arm is unique to zanidatamab and absent from the comparators.
- **Variance:** vs trastuzumab; trastuzumab + pertuzumab; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Efficacy shown in gastric models, not biliary; HER2 biology is shared but no CCA xenograft tested here. Mouse complement is impaired, so the CDC contribution measured in human serum may not carry into the in-vivo models.

—/— (2023)

Reference: — · **Type:** trial · **Inclusion:** included

- **Intervention:** Adagrasib (KRAS G12C covalent inhibitor) +/- combination
- **Indication / model:** KRAS G12C-mutated advanced solid tumors incl. biliary tract (2L+); KRAS G12C mutation
- **Design:** Open-label phase 1/2 basket (KRYSTAL-class adagrasib program)
- **Effect size:** — — Objective response rate; safety
- **Case fit:** partial
- **Notes:** (registration only — no peer-reviewed publication yet; toxicity table not extracted)

Subbiah/Wirth (2022) — Lancet Oncology

Reference: [PMID 36029780](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Selpercatinib
- **Indication / model:** RET fusion-positive solid tumors other than lung/thyroid (basket) (2L+); RET fusion-positive

non-lung/non-thyroid tumors; 58% had refractory GI malignancies. Biliary patients enroll on a confirmed RET fusion

- **Design:** Open-label phase 1/2 basket (LIBRETTO-001), tumour-agnostic cohort
- **n:** 45
- **Effect size:** 43.9 % — ORR (independent review)
- **Variance:** 95% CI 28.5–60.3
- **Toxicities:** Grade 3+ AE (any) $G \geq 3$ (n=45, 56%); Hypertension $G \geq 3$ (n=45, 22%); Increased ALT $G \geq 3$ (n=45, 16%); Increased AST $G \geq 3$ (n=45, 13%); Grade 5 TEAE G5 (3/45, 7%)
- **Case fit:** partial
- **Notes:** Tumor-agnostic RET-fusion basket covering non-lung/non-thyroid tumors; RET fusions are rare in CCA and need RNA-based fusion NGS. ORR is on 41 evaluable of 45 enrolled.

Lidsky/Lidsky (2022) — npj Precision Oncology

Reference: [PMID 36274097](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Pemigatinib + HDAC inhibitor (quisinostat)
- **Indication / model:** Patient-derived FGFR2-G3BP2 fusion intrahepatic CCA matched cell line, organoid, and xenograft (DUC18828)
- **Design:** A high-throughput screen of >2,100 compounds across the matched models flagged HDAC inhibition as the strongest partner for FGFR blockade. The combination cut proliferation and pushed cells into apoptosis beyond what FGFR inhibition managed alone.
- **n:** not reported per arm
- **Effect size:** moderate — The pemigatinib-quisinostat pair shrank PDX tumors by 59.4% versus 23.8% for pemigatinib alone and 15.1% for quisinostat alone, with matching gains in the organoid and cell line. This is one of the few preclinical platforms built from a fusion-positive intrahepatic CCA patient.
- **Variance:** vs vehicle, pemigatinib alone, quisinostat alone; translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Single-patient model with an FGFR2-G3BP2 fusion; the HDAC combination is investigational, not the on-label monotherapy the patient would receive first.

Wu/Bardeesy (2022) — Cancer Discovery

Reference: [PMID 34880079](#) · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Ivosidenib (AG-120)
- **Indication / model:** Alb-Cre Kras-G12D IDH1-R132C/H genetically engineered mouse model of intrahepatic CCA; syngeneic allografts from IDH1-R132C tumor lines, with NSG immunodeficient controls
- **Design:** Mutant IDH1 generates R-2-hydroxyglutarate, which blocks TET2 and the cell's interferon-gamma response, letting the tumor hide from immune attack. Inhibiting mutant IDH1 reactivates TET2, switches on viral-mimicry transposable elements and cGAS-STING, and restores IFN-gamma sensitivity.
- **n:** not reported per arm
- **Effect size:** strong — AG-120 dropped R-2HG by more than 85%, slowed syngeneic tumors, and extended survival, but did nothing in immunodeficient NSG mice carrying the same cells, so the antitumor effect runs through the immune system. The drug also synergized with CTLA-4 blockade, clearing tumors in most treated animals.
- **Variance:** vs vehicle; same cell lines in immunodeficient NSG mice; translatability: high

- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Mouse genetics pair IDH1-R132 with Kras-G12D, which is not the typical human IDH1-mutant CCA background; the immune-dependent mechanism may overstate single-agent benefit seen clinically in ClarIDHy.

Abou-Alfa/Vogel (2020) — Lancet Oncology

Reference: [PMID 32203698](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pemigatinib
- **Indication / model:** Previously treated advanced/metastatic cholangiocarcinoma with FGFR2 fusion or rearrangement (2L+); FGFR2 fusion/rearrangement cohort (cohort A), prior gemcitabine-based chemotherapy; ECOG 0-2
- **Design:** Single-arm, open-label, multicohort phase 2 (FIGHT-202)
- **n:** 107
- **Effect size:** 35.5 % — ORR (independent central review)
- **Variance:** 95% CI 26.5–45.4
- **Toxicities:** Treatment-related grade 3+ AE (any) $G \geq 3$ (n=146, 64%); Hyperphosphatemia (81/146, 55%); Hypophosphatemia $G \geq 3$ (18/146, 12%); Alopecia (67/146, 46%); Dysgeusia (55/146, 38%); Nail toxicity (62/146, 42%); Arthralgia $G \geq 3$ (9/146, 6%); Stomatitis $G \geq 3$ (8/146, 5%); Palmar-plantar erythrodysesthesia (16/146, 11%); Serous retinal detachment (6/146, 4%)
- **Case fit:** partial
- **Notes:** Pivotal trial driving FDA accelerated approval of pemigatinib in FGFR2-fusion CCA. Gated on a confirmed FGFR2 fusion, untested here (ngs_pending); efficacy and safety pulled from PMC full text PMC8461541.

Abou-Alfa/Zhu (2020) — Lancet Oncology

Reference: [PMID 32416072](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Ivosidenib
- **Indication / model:** Previously treated advanced cholangiocarcinoma with IDH1 R132 mutation (2L+); IDH1 R132-mutant CCA after 1-2 prior regimens; crossover from placebo to ivosidenib permitted on progression
- **Design:** Randomized, double-blind, placebo-controlled phase 3 (ClarIDHy)
- **n:** 185
- **Effect size:** 0.37 HR — PFS HR vs placebo (independent review)
- **Variance:** 95% CI 0.25–0.54; $p < 0.0001$
- **Toxicities:** Ascites $G \geq 3$ (9/121, 7%); Fatigue $G \geq 3$ (4/121, 3%); Nausea $G \geq 3$ (3/121, 2%); QT prolongation G3 (1/121, 1%); Serious AE (any) (36/121, 30%)
- **Case fit:** partial
- **Notes:** Randomized phase 3 supporting FDA approval of ivosidenib in IDH1-mutant CCA. Gated on a confirmed IDH1 R132 mutation, untested here. Full efficacy/safety from PMC full text PMC7523268.

Subbiah/Wainberg (2020) — Lancet Oncology

Reference: [PMID 32818466](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Dabrafenib + trametinib

- **Indication / model:** BRAF V600E-mutated biliary tract cancer (2L+); Molecularly confirmed BRAF V600E (NGS/PCR; VE1 IHC suggestive but not sufficient); non-V600E class 2/3 BRAF alterations do not qualify
- **Design:** Open-label, single-arm phase 2 basket, biliary cohort (ROAR)
- **n:** 43
- **Effect size:** 47.0 % — ORR (investigator)
- **Variance:** 95% CI 31.0–62.0
- **Toxicities:** Serious AE (any) (17/43, 40%); Treatment-related serious AE (9/43, 21%); Pyrexia (8/43, 19%); Increased gamma-glutamyltransferase G $\geq$ 3 (5/43, 12%)
- **Case fit:** partial
- **Notes:** ROAR biliary cohort plus the tumor-agnostic BRAF V600E label give two routes to treatment. Gated on a molecularly confirmed BRAF V600E, untested here. Durability/OS from conference and OncLive follow-up.

Marabelle/Diaz (2020) — Journal of Clinical Oncology

Reference: [PMID 31682550](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab
- **Indication / model:** MSI-H/dMMR advanced noncolorectal solid tumors (incl. biliary) (2L+); MSI-H/dMMR by PCR or IHC; 27 tumor types, cholangiocarcinoma among the most common. No CCA-specific subgroup ORR published
- **Design:** Open-label, multicohort phase 2 basket (KEYNOTE-158)
- **n:** 233
- **Effect size:** 34.3 % — ORR (independent central review)
- **Variance:** 95% CI 28.3–40.8
- **Toxicities:** Treatment-related grade 3-5 AE (any) G $\geq$ 3 (34/233, 14.6%); Treatment-related AE (any) (151/233, 64.8%); Pneumonia G5 (1/233)
- **Case fit:** partial
- **Notes:** Tumor-agnostic basket admitting MSI-H/dMMR biliary cancer on the biomarker; MSI-H is rare in BTC (~1-3%) but decision-changing. No CCA-specific subgroup ORR reported in this publication.

Marabelle/Bennouna (2020) — Lancet Oncology

Reference: [PMID 32919526](#) · **Type:** clinical · **Inclusion:** included

- **Intervention:** Pembrolizumab
- **Indication / model:** TMB-high advanced solid tumors ($\geq$ 10 mut/Mb), incl. biliary (2L+); tTMB-high $\geq$ 10 mut/Mb on FoundationOne CDx; biliary admitted through the biomarker, not a dedicated cohort
- **Design:** Prospective biomarker analysis, phase 2 basket (KEYNOTE-158)
- **n:** 102
- **Effect size:** 29.0 % — ORR in tTMB-high subgroup
- **Variance:** 95% CI 21.0–39.0
- **Toxicities:** Safety mirrors the broader KEYNOTE-158 pembrolizumab population; this biomarker analysis did not report separate grade 3+ rates for the TMB-high subgroup.
- **Case fit:** partial
- **Notes:** TMB-high subgroup analysis supporting the tumor-agnostic TMB-high label; the same CGP that resolves the other axes returns this value. Toxicities[] empty because this biomarker-analysis publication reports no per-term safety for the subgroup (abstract and ASCO Post summary checked; safety deferred to the parent KEYNOTE-158 reports).

Doebele/Demetri (2020) — Lancet Oncology

Reference: [PMID 31838007](#) · Type: clinical · Inclusion: included

- **Intervention:** Entrectinib
- **Indication / model:** NTRK fusion-positive advanced solid tumors (integrated basket) (2L+); NTRK1/2/3 fusion across tumor types; biliary patients enroll on the fusion regardless of subsite
- **Design:** Integrated analysis of three phase 1-2 trials (ALKA-372-001, STARTRK-1, STARTRK-2)
- **n:** 54
- **Effect size:** 57.4 % — ORR (blinded independent review)
- **Variance:** 95% CI 43.2–70.8
- **Toxicities:** Treatment-related grade 3-4 AE (any) G_≥3 (n=193, 41.5%); Weight increase G_≥3 (n=193, 8.3%); Anemia G_≥3 (n=193, 5.2%); Fatigue G_≥3 (n=193, 4.7%); Dysgeusia (n=193, 35.2%)
- **Case fit:** partial
- **Notes:** Tumor-agnostic NTRK-fusion basket; rare in CCA (<1%), best caught by RNA-based fusion NGS. Efficacy from the 2020 Lancet Oncology paper; safety rates from the 2022 updated integrated analysis (PMC9365368) since the original per-term grade 3-4 table was not separately accessible.

Hong/Hyman (2020) — Lancet Oncology

Reference: [PMID 32092306](#) · Type: clinical · Inclusion: included

- **Intervention:** Larotrectinib
- **Indication / model:** TRK fusion-positive advanced solid tumors (pooled basket) (2L+); NTRK1/2/3 fusion across tumor types and ages; biliary patients treated on the fusion irrespective of subsite
- **Design:** Pooled analysis of three phase 1/2 trials (incl. NAVIGATE)
- **n:** 159
- **Effect size:** 79.0 % — ORR (independent review)
- **Variance:** 95% CI 72.0–85.0
- **Toxicities:** Grade 3 AE (any) G3 (33/260, 13%); Grade 4 AE (any) G4 (2/260, 1%); Increased ALT G_≥3 (n=260, 3%); Anemia G_≥3 (n=260, 2%); Decreased neutrophils G_≥3 (n=260, 2%)
- **Case fit:** partial
- **Notes:** Highly selective TRK inhibitor with a tumor-agnostic label and the most durable responses among the fusion inhibitors. Same rare-fusion gate as entrectinib. Safety from the 260-patient pooled safety population.

Liu/Wu (2020) — PLoS ONE

Reference: [PMID 32315352](#) · Type: preclinical · Inclusion: included

- **Intervention:** Pemigatinib (INCB054828)
- **Indication / model:** FGFR2-TACC fusion CCA patient-derived xenograft (CTG-0997, FGFR2-TRA2B fusion) plus a panel of FGFR1/2/3-altered cell-line xenografts
- **Design:** Selective ATP-competitive inhibitor of FGFR1/2/3 (IC₅₀ 0.4/0.5/1.0 nM, sparing FGFR4 at 30 nM). In an FGFR2-fusion tumor the fusion drives ligand-independent FGFR signaling, and the drug shuts that down.
- **n:** 12 tumors at the 1 mg/kg dose
- **Effect size:** strong — In the FGFR2-fusion CCA PDX, 10 of 12 tumors shrank at 1 mg/kg and both doses beat vehicle on growth inhibition (P<0.01). Adding cisplatin outperformed either agent alone.
- **Variance:** vs vehicle; cisplatin alone in the combination arm; translatability: high

- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** The CCA PDX carries an FGFR2-TRA2B fusion rather than the more common FGFR2-BICC1; the patient's fusion is untested (ngs_pending).

Hallin/Christensen (2020) — Cancer Discovery

Reference:PMID 31658955 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Adagrasib (MRTX849)
- **Indication / model:** Panel of KRAS G12C cell-line and patient-derived xenografts (NSCLC, colorectal, pancreatic); biliary not represented
- **Design:** Covalent inhibitor (IC50 ~5 nM) that locks the mutant cysteine-12 of GDP-bound KRAS G12C, holding the protein inactive and cutting off downstream RAS signaling.
- **n:** broad model panel, per-arm n not reported
- **Effect size:** strong — Adagrasib produced broad antitumor activity across KRAS G12C models, with complete responses in a subset, at tolerated doses. The work also mapped co-mutations and feedback reactivation that shape which tumors respond.
- **Variance:** vs vehicle; translatability: low
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** No biliary model in the panel; KRAS G12C is roughly 1% of CCA, so this is mechanism support for a rare candidate feature rather than a tissue-matched proof of concept.

Goyal/Bardeesy (2019) — Cancer Discovery

Reference:PMID 31109923 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Futibatinib (TAS-120)
- **Indication / model:** Patient-derived FGFR2-fusion intrahepatic CCA cells with acquired kinase-domain resistance mutations; serial biopsies and ctDNA from 4 patients pretreated with BGJ398 or Debio 1347
- **Design:** Irreversible covalent pan-FGFR inhibitor binding outside the ATP cleft, so several FGFR2 kinase-domain mutations that block reversible inhibitors (e.g. V564F, N550H) stay sensitive.
- **n:** 4 patients with paired patient-derived cells
- **Effect size:** strong — TAS-120 held activity against multiple FGFR2 resistance mutations that emerged on BGJ398 or Debio 1347, and the four patients who progressed on those reversible inhibitors responded to TAS-120. The mutation that resisted TAS-120 differed from the ones resisting reversible drugs, which points to a sequencing strategy.
- **Variance:** vs reversible ATP-competitive FGFR inhibitors (BGJ398, Debio 1347); translatability: high
- **Toxicities:** n/a (preclinical)
- **Case fit:** strong
- **Notes:** Mechanism-of-resistance study in fusion-positive intrahepatic CCA, so it matters only after FGFR2 confirmation and prior FGFR-inhibitor exposure. Heavy reliance on ctDNA and a small patient set.

Subbiah/Brandhuber (2018) — Annals of Oncology

Reference:PMID 29912274 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Selpercatinib (LOXO-292)
- **Indication / model:** CCDC6-RET non-small-cell lung and RET-C634W medullary thyroid cell-line xenografts plus two CCDC6-RET patient-derived xenografts
- **Design:** Highly selective ATP-competitive RET inhibitor (wild-type IC₅₀ 0.92 nM, >250-fold selective over most other kinases) that blocks fusion-driven and mutant RET signaling, including the V804 gatekeeper variants.
- **n:** not reported per arm
- **Effect size:** strong — Selpercatinib gave dose-dependent tumor growth inhibition exceeding 90% and drove complete regressions in several RET fusion or mutant models at 3-10 mg/kg. Selectivity against non-RET kinases predicts the clean off-target profile seen clinically.
- **Variance:** vs vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Lung and thyroid RET models, no biliary tissue; relevance to CCA is the tumor-agnostic RET-fusion mechanism, and RET fusions are rare in biliary tract cancer.

Le/Diaz (2017) — Science

Reference:PMID 28596308 · **Type:** preclinical · **Inclusion:** excluded — The predictive axis here is host neoantigen load from mismatch-repair deficiency or high TMB, not a tumor-cell drug target that preclinical mouse or organoid work can model for this patient. The mechanism evidence for anti-PD-1 in dMMR tumors is human (Le et al.) rather than a CCA-specific preclinical proof of concept, so it backs the biomarker but adds nothing to a targetable-feature preclinical dossier.

- **Intervention:** Pembrolizumab (MSI-H/dMMR, TMB-high)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: The predictive axis here is host neoantigen load from mismatch-repair deficiency or high TMB, not a tumor-cell drug target that preclinical mouse or organoid work can model for this patient. The mechanism evidence for anti-PD-1 in dMMR tumors is human (Le et al.) rather than a CCA-specific preclinical proof of concept, so it backs the biomarker but adds nothing to a targetable-feature preclinical dossier.

Ogitani/Agatsuma (2016) — Cancer Science

Reference:PMID 27166974 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Trastuzumab deruxtecan (DS-8201a)
- **Indication / model:** Admixed HER2-high (3+) and HER2-negative tumor-cell xenografts modeling HER2 heterogeneity; NCI-N87 and HER2-variable lines
- **Design:** HER2 antibody linked to a membrane-permeable topoisomerase I inhibitor (DXd) at a drug-to-antibody ratio of 8. After internalization and cathepsin cleavage, the payload diffuses out and kills neighboring cells regardless of their own HER2 level.
- **n:** not reported per arm
- **Effect size:** moderate — DS-8201a killed HER2-negative cells when they were mixed with HER2-positive cells, an effect T-DM1 did not produce. This bystander killing explains activity across heterogeneous and lower-HER2 tumors, which fits the variable HER2 staining seen in biliary tract cancer.
- **Variance:** vs T-DM1 (trastuzumab emtansine); HER2-negative cells alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Bystander mechanism shown in breast/gastric-type models, not biliary; relevance to CCA rests on the

IHC-driven DESTINY-PanTumor02 signal rather than a CCA xenograft.

Ardini/Galvani (2016) — Molecular Cancer Therapeutics

Reference:PMID 26939704 · **Type:** preclinical · **Inclusion:** excluded — Entrectinib preclinical work covers the same TRK-fusion mechanism in colorectal (KM12) and neuroblastoma models already captured by the larotrectinib row, with no biliary tissue. For the NTRK candidate feature it duplicates rather than extends the included TRK proof of concept.

- **Intervention:** Entrectinib (RXDX-101)
- **Toxicities:** n/a (preclinical)
- **Notes:** Excluded: Entrectinib preclinical work covers the same TRK-fusion mechanism in colorectal (KM12) and neuroblastoma models already captured by the larotrectinib row, with no biliary tissue. For the NTRK candidate feature it duplicates rather than extends the included TRK proof of concept.

Doebele/Shah (2015) — Cancer Discovery

Reference:PMID 26216294 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Larotrectinib (LOXO-101)
- **Indication / model:** KM12 colorectal cells carrying a TPM3-NTRK1 fusion; subcutaneous KM12 xenografts and a patient-derived sarcoma model
- **Design:** Selective TRK inhibitor (IC50 3.5 nM against TPM3-TRKA in KM12) that shuts down fusion-driven TRK autophosphorylation and the downstream AKT and ERK signaling the fusion turns on.
- **n:** n=10 mice/group (KM12 xenograft)
- **Effect size:** strong — Larotrectinib stopped KM12 xenograft growth and dropped pTRK, pAKT, and pERK, with activity holding across TPM3-NTRK1, MPRIP-NTRK1, TRIM24-NTRK2, and ETV6-NTRK3 variants. The matched patient with an NTRK fusion sarcoma responded.
- **Variance:** vs diluent vehicle; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Colorectal and sarcoma models, not biliary; the case rests on the tumor-agnostic fusion mechanism since NTRK fusions are under 1% in CCA.

King/Laquerre (2013) — PLoS ONE

Reference:PMID 23844038 · **Type:** preclinical · **Inclusion:** included

- **Intervention:** Dabrafenib + trametinib
- **Indication / model:** BRAF V600E A375P melanoma and Colo205 colorectal xenografts; rat skin model for paradoxical lesions
- **Design:** Dabrafenib blocks mutant BRAF and trametinib blocks MEK downstream, so the combination hits two nodes of the MAPK cascade. Doubling up suppresses the MEK-driven pathway reactivation that lets tumors escape single-agent BRAF inhibition.
- **n:** not reported per arm
- **Effect size:** strong — The combination beat either drug alone for tumor growth control in both V600E xenografts and prevented the hyperproliferative skin lesions that single-agent BRAF inhibition caused in rats. That dual benefit, efficacy plus less paradoxical toxicity, is the rationale for combining the two clinically.

- **Variance:** vs vehicle; dabrafenib alone; trametinib alone; translatability: med
- **Toxicities:** n/a (preclinical)
- **Case fit:** cross_tumor_only
- **Notes:** Melanoma and colorectal models, no biliary tissue; BRAF V600E signaling is shared but CCA-specific preclinical data are sparse. Supports the tumor-agnostic mechanism behind the ROAR biliary cohort.